



EARTH'S MOON

KEY QUESTIONS

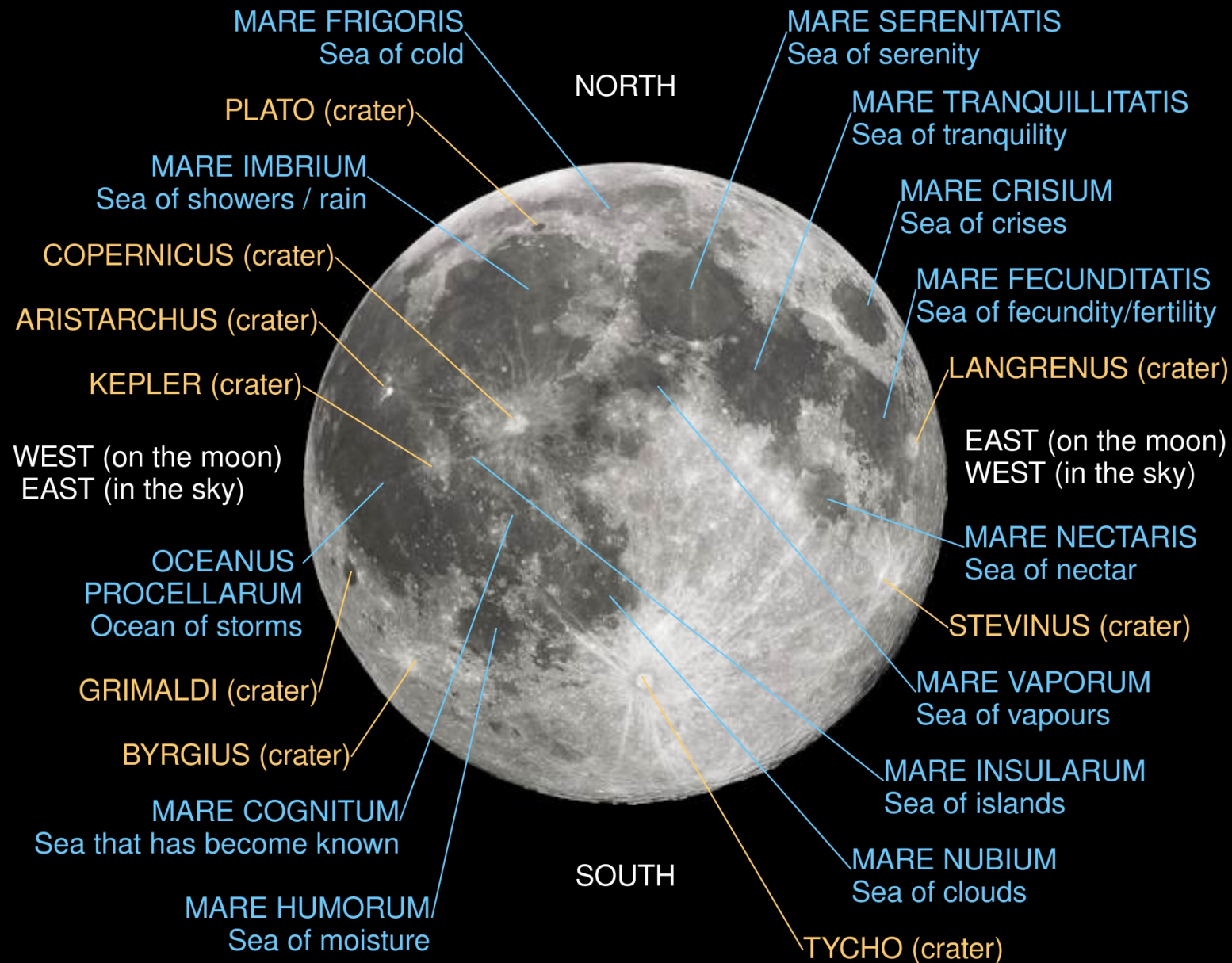
- What are the moon's characteristics?
- How did the moon form?
- How does the moon move?
- Why does the moon have phases?
- What does the moon have to do with tides?
- Why is the moon important to us?



CHARACTERISTICS

- Earth's only natural **satellite**
- formed ~50 million years after the solar system
- mass = 1.2% of Earth
- Size is about $\frac{1}{4}$ of Earth
- Surface Gravity is about $\frac{1}{6}$ of Earth
- Larger than all known dwarf planets

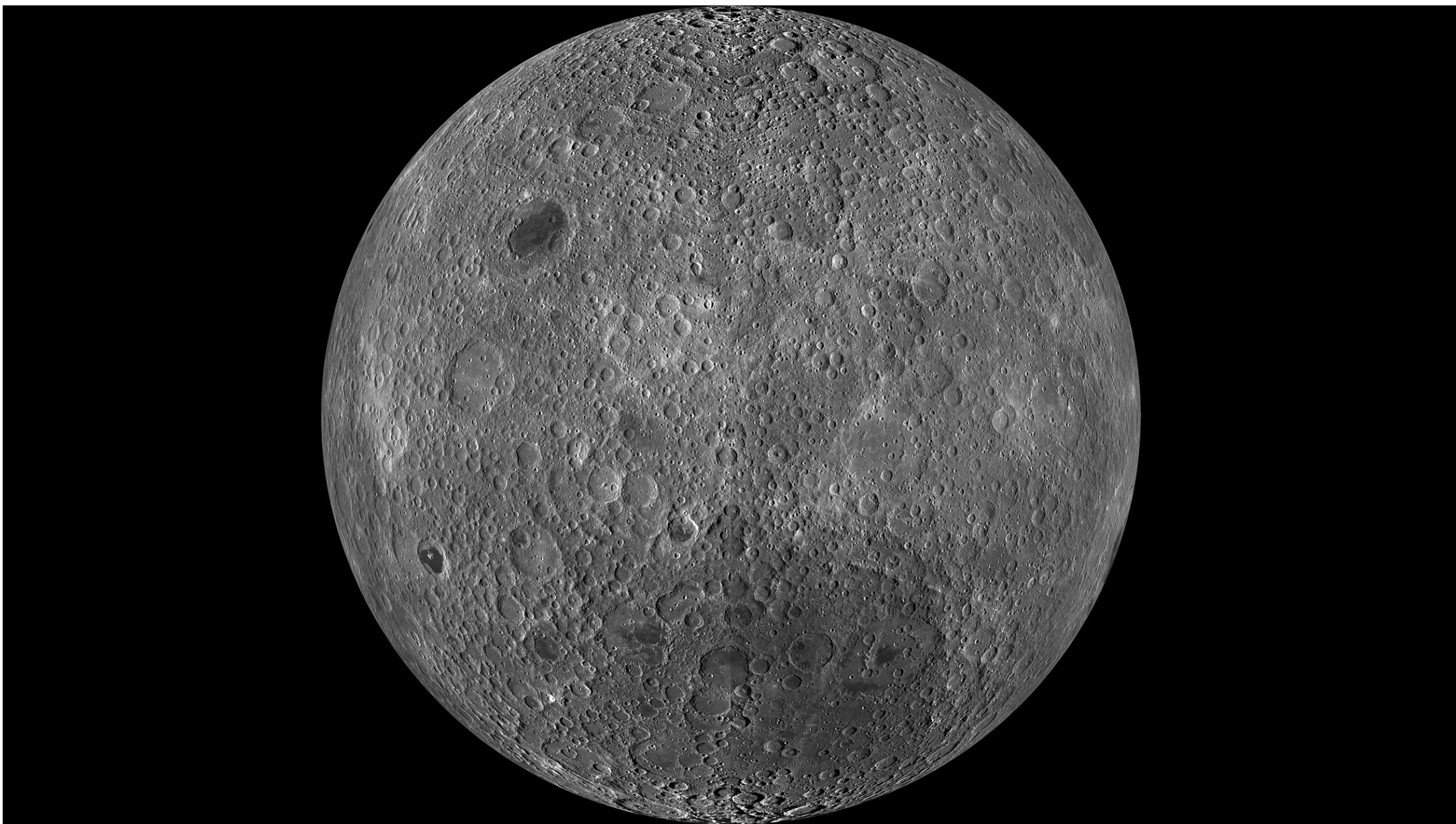




CHARACTERISTICS

- Temperatures: 120 °C to -171 °C
- Large dark spots on near side called “Maria” or “Seas”
- Far side has many craters and has been studied less
- Volcanos that went extinct 1,200,000,000 years ago
- Covered in fine dust that sticks and can damage lungs
- Little to no atmosphere and magnetosphere





FORMATION:

- We are still learning about how the Moon formed!
- The isotopes of elements found on the Moon match Earth very closely, but the Moon has fewer metals, which is still a bit of a mystery.
- The Giant Impact Hypothesis is the leading theory today. It states that when the Earth was still forming, a Mars-sized object (called Theia) hit Earth from the side and the ejected debris formed the Moon.

Spaceside crust (40 mi)

Aristarchus crater

Cold sea

Plato crater

Rocky mantle

Partially molten zone

Iron core

Rainy sea

Sea of Serenity

Sea of Tranquility

Sea of Vapors

Ocean of storms

Earthside crust (31 mi)

Sea of Islands

Eastern sea

Grimaldi crater

Familiar Sea

Sea of Moisture

Sea of Clouds

Byrgius crater

Rotational pole

Ecliptic

Orbit

Craters

Copernicus

Kepler

Tycho

To the Sun

(waning gibbous)

To Earth

Orbit of the Moon

The Moon

All features drawn to scale

Field of view 49.134°
x shift + 0.120
y shift + 0.040

DISCUSSION:

- If you stood in one place on the Moon, would you experience day and night?
- How long would a day last on the Moon?

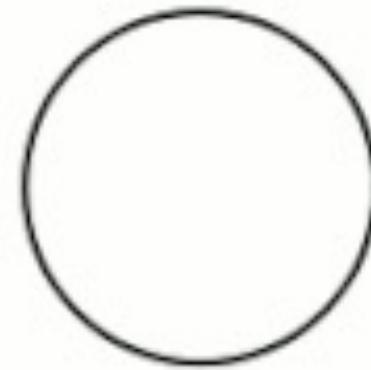
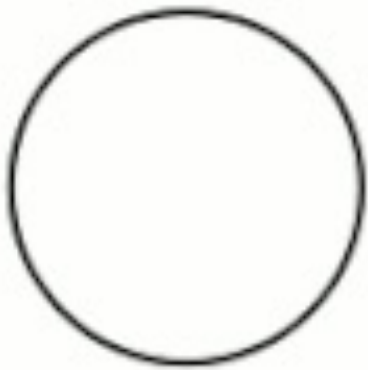


MOVEMENT:

- The Moon rotates once every 29.5 Earth days (1 month)
- Both sides experience the same amount of sunlight
- The Moon is tidally locked with Earth, meaning the same side always faces Earth.
- The Moon's orbit is growing further from the Earth at a rate of 38mm per year, about the rate of fingernail growth
- This lengthens Earth's day by 17 microseconds every year.



TIDAL LOCKING



Tidally locked:
Same side always faces planet

Not tidally locked

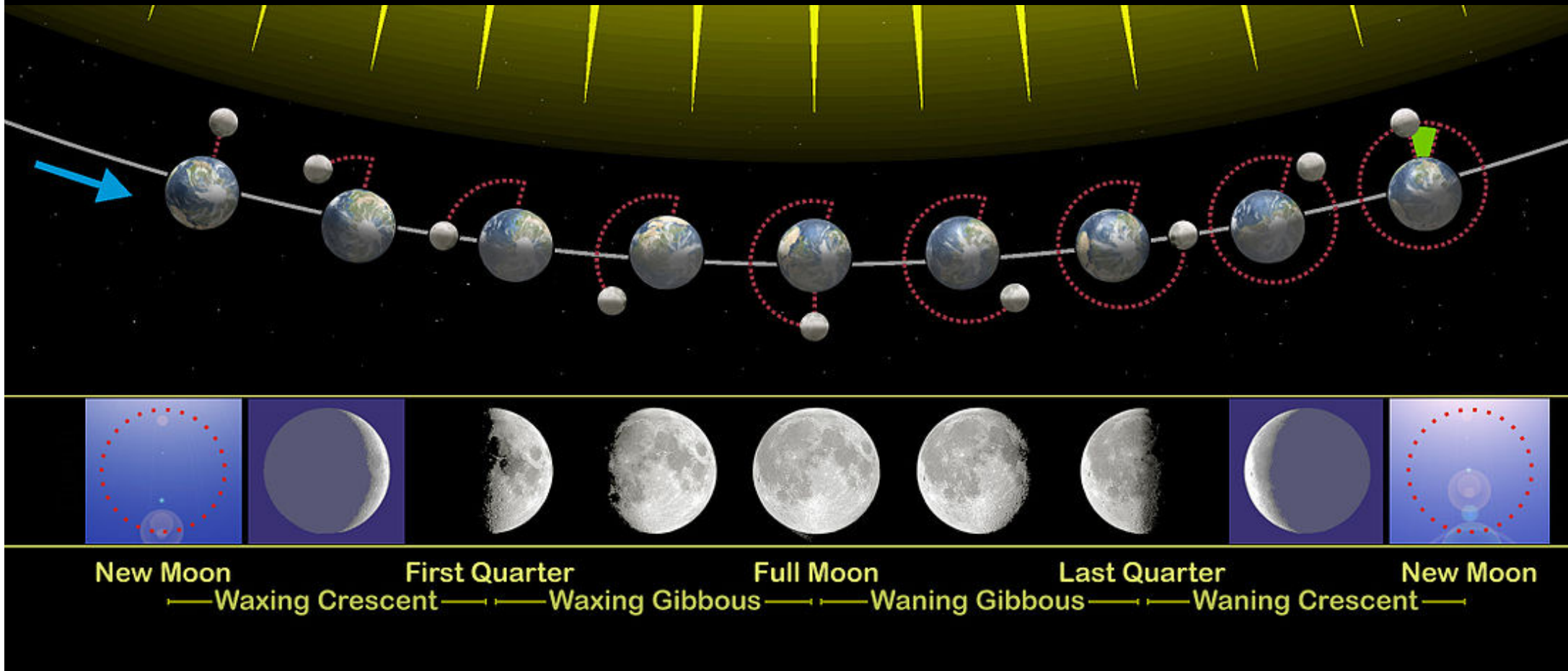
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PHASES:

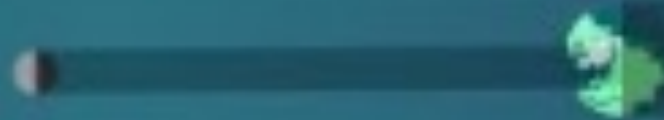
- The amount of surface that is visible and illuminated by the Sun changes as the Moon orbits Earth



PHASES:



Why don't we have a solar eclipse every month?



not to scale

WHAT ARE TIDES?

Eye on Acadia Wed Jul 15, 2020 09:18:24 AM

TIDES:

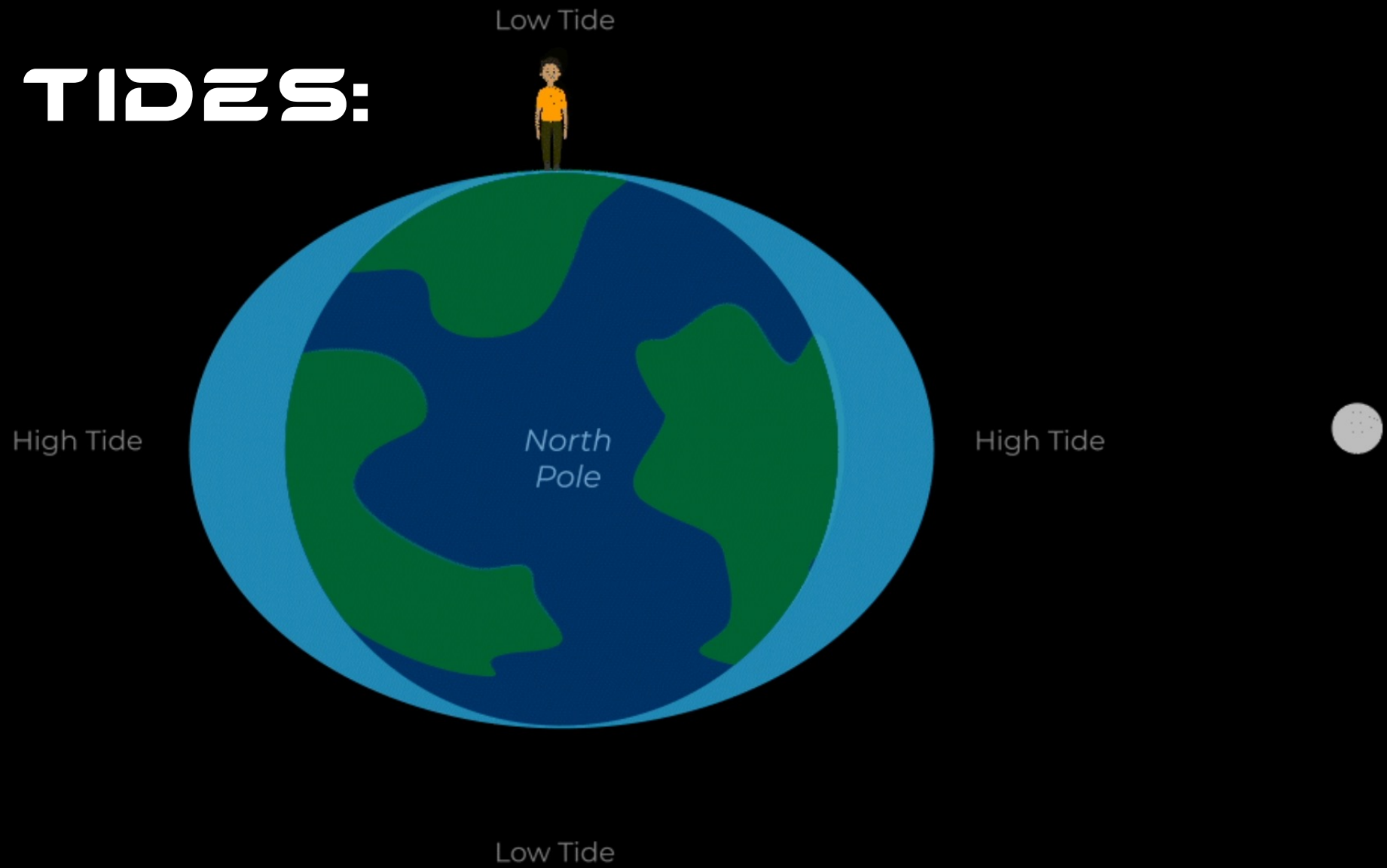
- Earth's oceans rise and fall at regular intervals
- Even Isaac Newton could not figure out why!
- French scientist Pierre-Simon Laplace wrote accurate equations in 1775, but the theory was not complete until 1921, when more information about the Moon was available



TIDES:



TIDES:

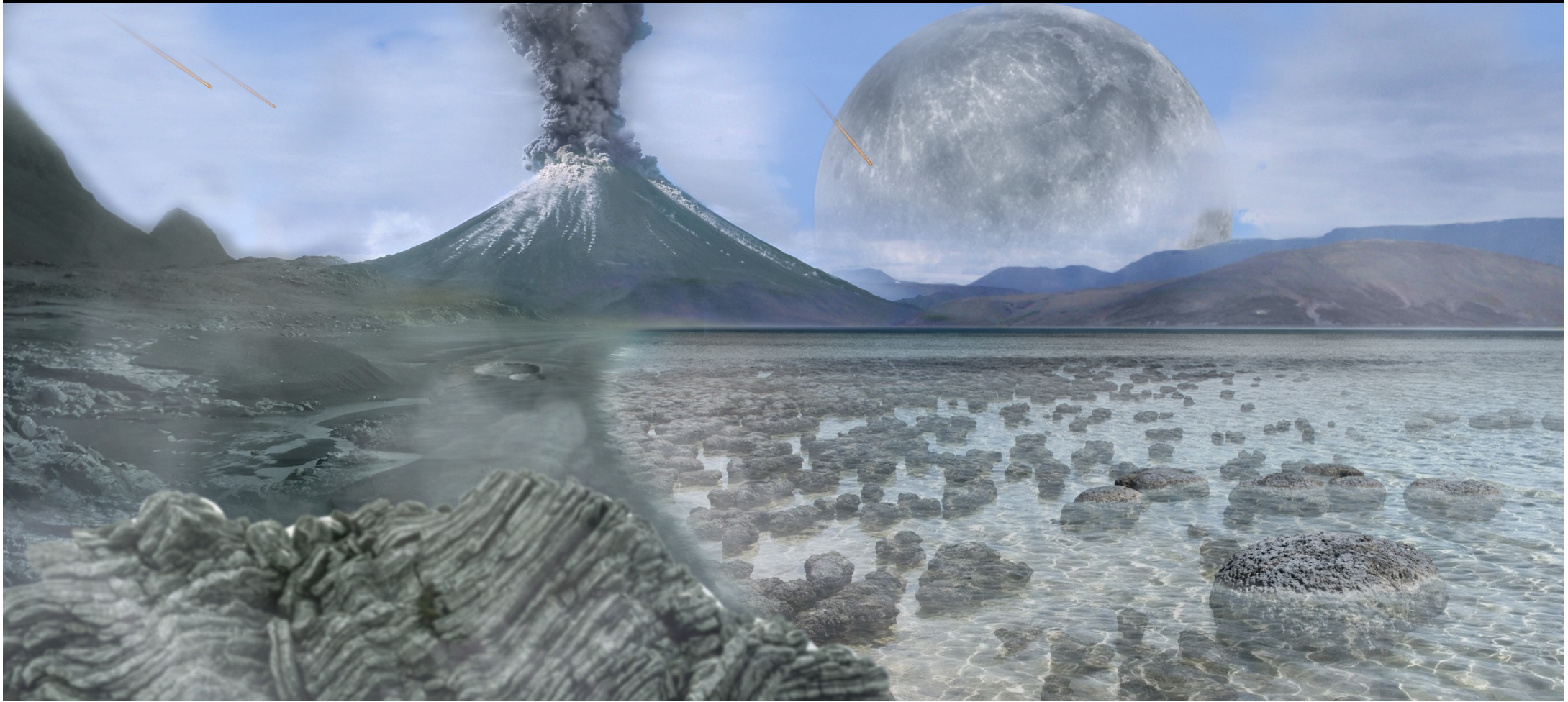


THE MOON AND HUMANS

- Cultural stories and wisdom passed down through generations
- Important for ancient calendars
- Teaches scientists about physics and other worlds
- Used by many animals to navigate at night
- Protects Earth from asteroids
- Helps determine the length of Earth's day
- Mainly responsible for the tides (the Sun also plays a role)



4 BILLION YEARS AGO



SOME PSEUDOSCIENCE

Since ancient times, people have claimed that the full moon has an effect on human behavior.

Many studies have been conducted to measure the effect of the full moon on accidents, violence, insanity, and other outcomes, but no study has supported a correlation with the lunar cycle.

The effect of gravity from the moon is too small to affect the water stored inside a person's body.



LUNAR EXPLORATION

- Soviet Luna probes began in 1959
- Apollo 11 mission lands on the moon in 1969
- Japan's Hiten probe 1990
- European Space Agency SMART-1 probe 2003
- Chinese probe Chang'e 1 probe 2007
- India's Chandrayaan-1 probe 2008
- American LRO orbiter 2009
- South Korean Danuri orbiter 2023

